

LECTURE 3: MATH1061/7861
TOPICS: LOGICAL EQUIVALENCE
DATE: THURSDAY, FEB. 26, 2026

Learning objectives:

- (1) Use a truth table to establish logical equivalence of two statements.
- (2) Apply logical equivalence laws to establish logical equivalence of two statements.

* Make sure to bring pen/paper
or ipads, tablets, ...

* Applied Classes start next week.

You must submit an exercise sheet
in this class.

This is an important part of your grade.

* Cannot be deferred to another week.

Two statements are said to be **logically equivalent** if their values are the same regardless of the values of the variables ($p, q, r, \dots$).

Q1: Let p and q be statement variables. Write down the truth tables for the following statements and determine whether they are logically equivalent.

Equivalence.

De Morgan's Law

$$\sim(p \wedge q),$$

$$\equiv$$

$$(\sim p) \vee (\sim q).$$

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$(\sim p) \vee (\sim q)$	
T	T	T	F	F	F	F	✓
T	F	F	T	F	T	T	✓
F	T	F	T	T	F	T	✓
F	F	F	T	T	T	T	✓

Logical Equivalences

Given any statement variables p, q and r , a tautology t and a contradiction c , the following logical equivalences hold.

Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge t \equiv p$	$p \vee c \equiv p$
Negation laws	$p \vee \sim p \equiv t$	$p \wedge \sim p \equiv c$
Double negative laws	$\sim(\sim p) \equiv p$	
Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
Universal bound laws	$p \vee t \equiv t$	$p \wedge c \equiv c$
De Morgan's laws	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
Absorption laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negations of t and c	$\sim t \equiv c$	$\sim c \equiv t$

Exercise: Pick one or two of these.

And show it is correct by drawing the truth table.

t = tautology = statement that is always true.

c = contradiction = Always false.

Note: $\sim$ takes precedent over $\wedge$ & $\vee$.

e.g. $\sim p \wedge q$ means $(\sim p) \wedge q$.

Q2:

Determine whether the following statement is true or false:

$$\sim p \wedge (p \wedge q) \equiv \sim p \vee q$$

Multiple ways to answer this. Two possible answers include:

- Via truth table:

p	q	$\sim p$	$p \wedge q$	$\sim p \wedge (p \wedge q)$	$\sim p \vee q$
T	T	F	T	F	T
T	F	F	F	F	F
F	T	T	F	F	T
F	F	T	F	F	T

↔
Not the same

The statement is false

- Via a counterexample

Let p be true and q be true.

Then the left side is $\sim p \wedge (p \wedge q) \equiv F \wedge (T \wedge T) \equiv F$

and the right side is $\sim p \vee q \equiv F \vee T \equiv T$.

These are not the same, so the statement is false.

Q3:

Use the laws of logical equivalence to prove that

$$(p \wedge \sim q) \wedge (\sim p \vee q)$$

is a contradiction.

By Negation Law,

First Approach It is enough to show $p \wedge \sim q$ is opposite of $\sim p \vee q$.

Indeed, by De Morgan ~~$\sim(p \wedge \sim q) \equiv \sim p \vee \sim(\sim q)$~~

$$\sim(p \wedge \sim q) \equiv (\sim p) \vee \sim(\sim q)$$

Double Negative Law $\Rightarrow \equiv \sim p \vee q$

Second Approach

De Morgan

$$(p \wedge \sim q) \wedge (\sim p \vee q) \equiv \underbrace{\sim(\sim p \vee q)}_{\sim r} \wedge \underbrace{(\sim p \vee q)}_r$$

Negation Law.

$$\Rightarrow \equiv \perp$$